

PROBLEM-2

To determine whether h_2 and h_3 perform worse than h_1 , we compare their 95% confidence intervals for true means.

Given

Test set size: $n=45$

Error rates:

$$h_1 = 6.67\% = 0.0667$$

$$h_2 = 8.89\% = 0.0889$$

$$h_3 = 13.3\% = 0.133$$

Confidence level = 95% $z = 1.96$



→ Confidence Interval for h_1
(0, 0.1398)

→ Confidence Interval for h_2

Standard Error:

$$= \sqrt{\frac{0.0889(1-0.0889)}{45}}$$

$$= \sqrt{\frac{0.0889 \times 0.9111}{45}}$$

$$= \sqrt{\frac{0.0809}{45}} = \sqrt{0.00180} = 0.0424$$

Margin of error

$$1.96 \times 0.0424 = 0.0831$$

confidence interval

$$0.0889 \pm 0.0831$$

$$= (0.0058, 0.1720)$$

→ Confidence Interval for h_3

Standard Error:

$$\sqrt{\frac{0.133(1-0.133)}{45}} = \sqrt{\frac{0.133(0.867)}{45}}$$



$$= \sqrt{\frac{0.1153}{45}} = \sqrt{0.00256} \approx 0.0506$$

Margin of error

$$1.96 \times 0.0506 = 0.0992$$

Confidence Interval

$$0.133 \pm 0.0992$$

$$= (0.0338, 0.2322)$$

for h_2 and h_1 ,

the intervals overlap significantly. Therefore, we cannot confidently say that h_2 performs worse than h_1 .

for h_3 and h_1 ,

the interval of h_3 is shifted more towards higher error values.

we have higher confidence that h_3 performs worse than h_1 , but it is still not statistically certain at 95% level.